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Shortcuts and Temporal Leakage in the Bangalore EEG Epilepsy Dataset

A Provenance Audit and Leakage-Aware Benchmark of Lightweight Seizure-Type Classifiers

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1. Introduction: Motivation

- Epilepsy affects about 50 million people worldwide [4].
- Visual EEG review is expert-intensive and slow.
- Automated seizure-type classification could support clinicians.
- Public benchmarks like BEED [1] drive models; their validity matters.

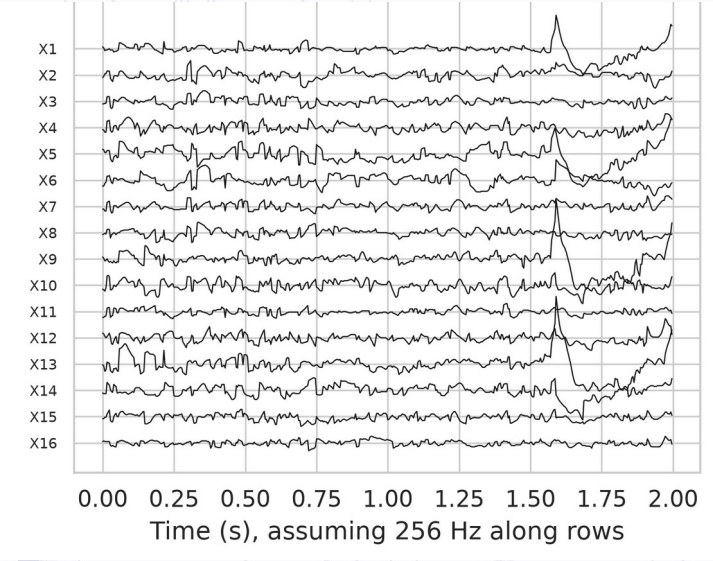


Fig. 1. BEED focal-seizure class: 512 consecutive rows drawn as 16 traces [1].

1. Introduction: The BEED Dataset

8,000 rows · 16 values per row (X1–X16) · documented: 16 channels, 256 Hz [1]

0 · Healthy

2,000 rows; documented as healthy adult subjects [1].

1 · Generalised seizures

2,000 rows; seizures involving both hemispheres.

2 · Focal seizures

2,000 rows; onset in a restricted brain region.

3 · Seizure events

2,000 rows; events with blinking or staring behaviours.

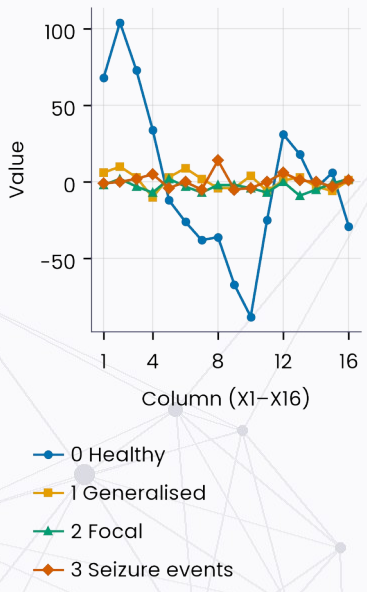


Fig. 2. One BEED row per class; healthy rows show far larger amplitude.

1. Introduction: Research Questions

RQ1 What do BEED rows and columns represent, and is this the same across classes?

RQ2 How much performance survives evaluation without temporal leakage?

RQ3 Once shortcuts are controlled, what separates the three seizure types?

RQ4 What is the computational footprint on a general-purpose CPU?

Random row split

Standard practice in prior BEED studies.

Adjacent rows can land in train and test.

Leakage-safe block split

Blocked CV: 200-row blocks.

Rolling origin: five 20% test blocks.

50-row purge gaps on both sides.

2. Dataset Audit: Where the Healthy Class Comes From

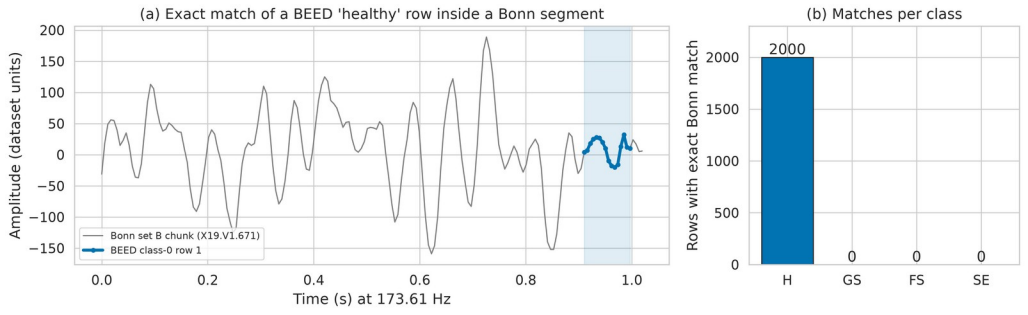


Fig. 3. A BEED healthy row identical to a Bonn set-B window (a); matches per class (b).

Match — All 2,000 healthy rows are identical to UCI windows [3], all at samples 159–174; classes 1–3: none.

Source — Bonn set B, eyes closed, 173.61 Hz [2]; classes 1–3: 256 Hz.

UCI page — Describes 80 × 200 = 16,000 segments, binary y; file: 8,000 rows, 4 classes, no subject IDs.

Effect — A single smoothness threshold separates healthy rows (accuracy 0.997).

2. Dataset Audit: Temporal Leakage

Consecutive seizure-class rows are adjacent time points, hence near-duplicates.

Random split 88–95% of seizure test rows have their neighbouring row in training.

Implication Random-split scores measure interpolation, not generalisation.

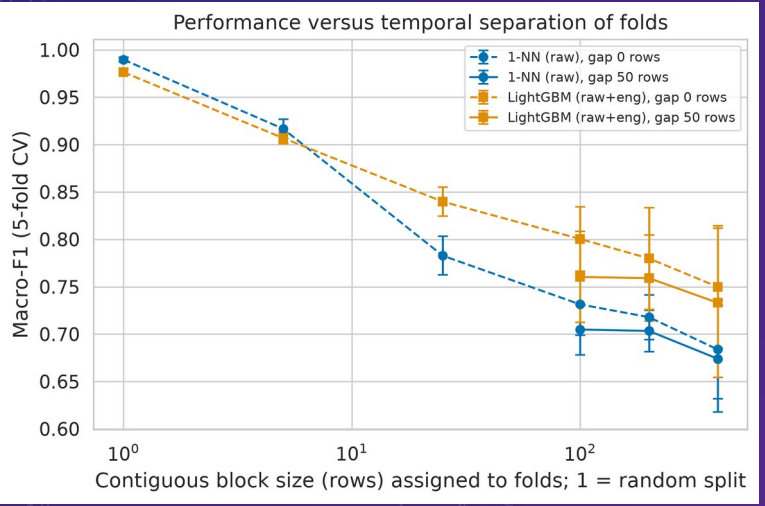


Fig. 4. Macro-F1 falls as folds use longer row blocks.

3. Methodology

Macro-F1 throughout; model pre-specified by blocked CV on development data

Splits

Random 80/20 vs blocked 5×2 CV with 200-row blocks and 50-row purge gaps.

Models

Primary: LightGBM (raw). Also LR, kNN, SVM-RBF, RF, XGBoost, four deep nets.

Statistics

Cluster-bootstrap CIs; corrected t-tests, Holm; Friedman/Nemenyi descriptive (anti-conservative).

Estimates

Primary: five-fold rolling origin. Secondary: final-20% hold-out.

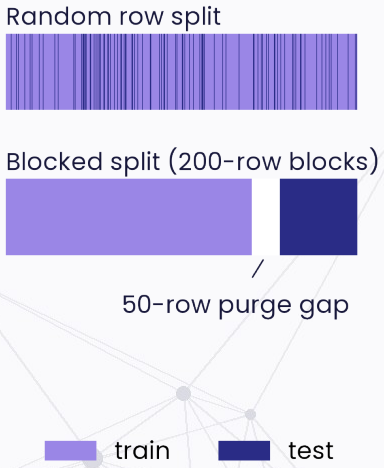


Fig. 5. Schematic: random versus blocked hold-out with a purge gap.

4. Results: Random vs Leakage-Safe Split

	Random	Leakage-safe
4-class macro-F1	0.98	0.75 ± 0.08
3-class macro-F1	0.98	0.68 ± 0.10
Single hold-out	–	0.81 / 0.75

Random: best model (kNN, raw), random-split CV. Leakage-safe: LightGBM (raw), chosen by blocked CV; five-fold rolling origin, mean ± SD. Single hold-out: final 20%, secondary (4-class / 3-class). 3-class: seizure types only.

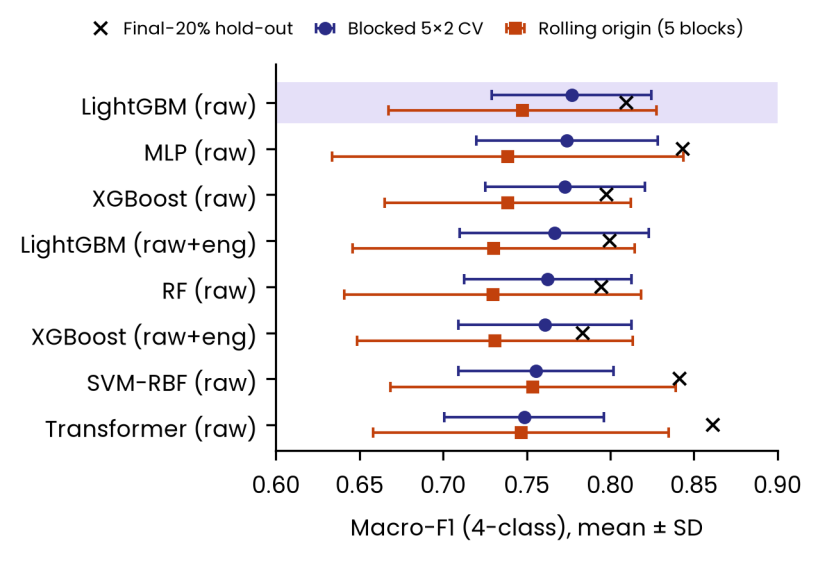


Fig. 6. Four-class leakage-safe macro-F1, top eight models by blocked CV; shaded: primary model.

4. Results: What the Models Look At

Leakage-safe feature importance

- Seizure types: a per-channel spatial amplitude pattern (X11, X8, X15); see slide 11.
- Healthy class: smoothness, e.g. Hjorth mobility.

Healthy class: a recording-source shortcut

- Healthy vs seizure: macro-F1 ≥ 0.99 for every model (best 0.999–1.000).
- One lag-1 autocorrelation threshold: accuracy 0.997.

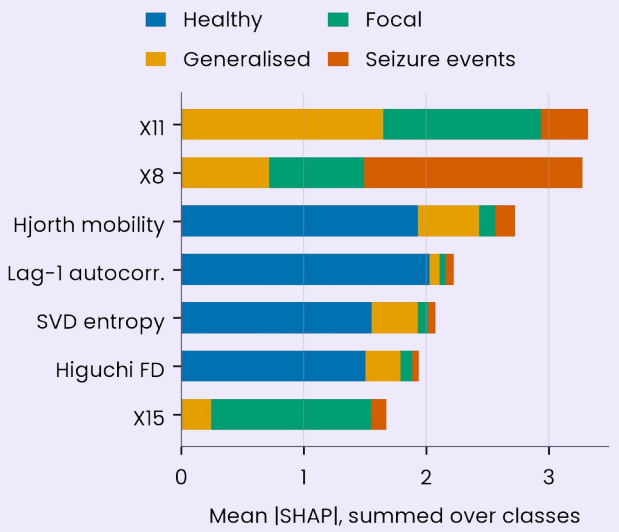


Fig. 7. Top SHAP features, LightGBM, four classes, leakage-safe split.

4. Results:
A Recording Fingerprint

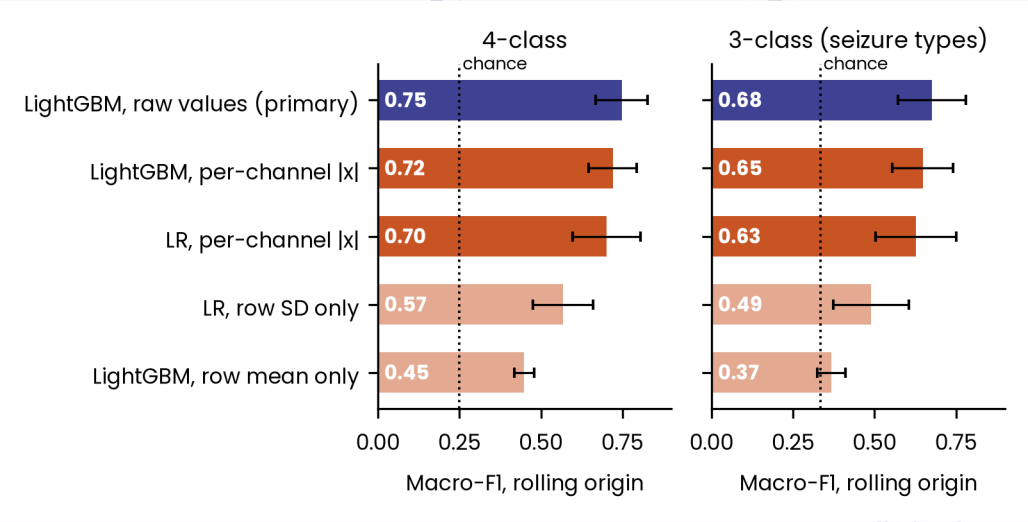


Fig. 8. Rolling-origin macro-F1 (mean ± SD) of the primary model and amplitude-only baselines.

Feature — Per-channel |x| only: 16 sign-free amplitudes, LightGBM.

4-class — 0.72 ± 0.07 vs 0.75 ± 0.08 (full model).

3-class — 0.65 ± 0.09 vs 0.68 ± 0.10 (full model).

Test — About 90% of above-chance score. Primary vs LR on |x|: +0.05, Holm p = 0.15–0.16, not significant.

4. Results: Computational Footprint (Server CPU)

Model (input)	Parameters	ONNX size	ms / sample	Macro-F1
MLP, tuned (raw 16)	3.5k	16 kB	0.05	0.77 (CV)
LightGBM, tuned (raw+eng., 73)	28.7k nodes	1.06 MB	3.97†	0.799 (hold-out)

Single thread, Intel Xeon server CPU (8-vCPU VM), Python; median of 500 single-sample calls; not edge hardware.
†Includes 3.48 ms feature extraction.

- LightGBM row: the tuned raw+engineered-features model (73 features), not the primary raw-value model; 0.799 is a single hold-out.
- MLP 0.77 is its tuned blocked 5×2 CV macro-F1 (0.774 ± 0.054).
- LightGBM inference alone: 0.50 ms (0.02 ms in ONNX Runtime); about 87% of its cost is feature extraction.

5. Discussion: Limitations

What remains valid

Leakage-safe scores estimate extrapolation to unseen time within the same recordings, not seizure-type generalisation.

Data

- No subject IDs, so no patient-level test.
- Recording identity and seizure type likely confounded (fingerprint result).
- Healthy class from Bonn set B; single dataset, no external test.

Methods

- CHB-MIT search for classes 1–3 found no match but was partial: 246 of 686 files, all 24 subjects; best 0.42 vs 0.80 threshold [5].
- Friedman/Nemenyi anti-conservative; deep models tuned on one inner hold-out; no edge hardware.

6. Conclusion: Takeaways and Future Work

Takeaways

- Random split 0.98 vs leakage-safe 0.75 ± 0.08 (3-class: 0.98 vs 0.68 ± 0.10).
- Per-channel |x| alone reaches 0.72 / 0.65: mostly a recording fingerprint.
- Healthy rows identical to Bonn set B windows [2], [3]: a source shortcut.
- Report pre-specified, leakage-safe (rolling-origin) estimates for BEED.

Future work

- Subject and recording metadata from the dataset creators.
- Validation on independent public EEG data.
- Same-source healthy recordings.
- Wider source search: TUSZ, Siena, filtered derivatives.

References

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Complete reference list: see the full paper.

THANK YOU

FOR YOUR ATTENTION

Questions?

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